



Loci

A **LOCUS (PLURAL LOCI)** IS THE PATH FOLLOWED BY A POINT THAT ADHERES TO A GIVEN RULE WHEN IT MOVES.

SEE ALSO

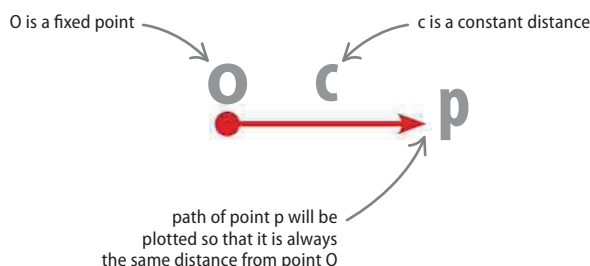
◀ 82–83 Tools in geometry

◀ 106–107 Scale drawings

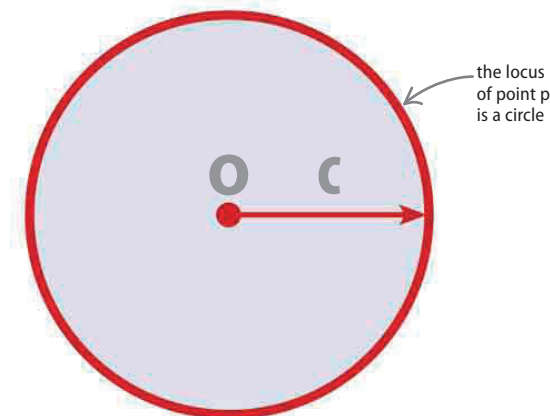
◀ 110–113 Constructions

What is a locus?

Many familiar shapes, such as circles and straight lines, are examples of loci because they are paths of points that conform to specific conditions. Loci can also produce more complicated shapes. They are often used to solve practical problems, for example pinpointing an exact location.



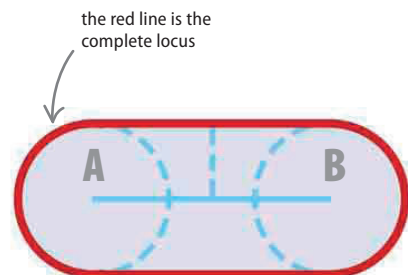
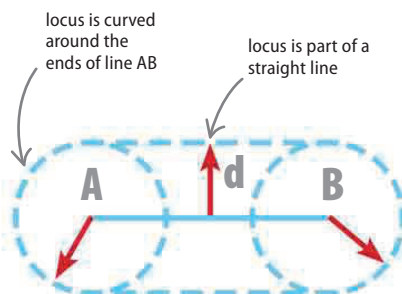
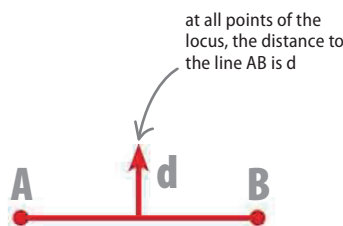
A **compass and a pencil** are needed to construct this locus. The point of the compass is held in the fixed point, O. The arms of the compass are spread so that the distance between its arms is the constant distance, c.



The **shape drawn** when turning the compass a full rotation reveals that the locus is a circle. The center of the circle is O, and the radius is the fixed distance between the compass point and the pencil (c).

Working with loci

To draw a locus it is necessary to find all the points that conform to the rule that has been specified. This will require a compass, a pencil, and a ruler. This example shows how to find the locus of a point that moves so that its distance from a fixed line AB is always the same.



Draw the line segment AB. A and B are fixed points. Now, plot the distance of d from the line AB.

Between points A and B, the locus is a straight line. At the end of these lines, the locus is a semicircle. Use a compass to draw these.

This is the completed locus. It has the shape of a typical athletics track.